G. Details and Results on Transformer Arithmetic Experiments

Charton (2023) trains sequence-to-sequence transformers to predict the greatest common divisor (GCD) of two positive integers, encoded as sequences of digits in some base B. He observes that model predictions are deterministic: for any pair (a, b) with GCD k, the model predicts a single value $f(k)$. Predictions are correct (i.e. $f(k) = k$) when the GCD is a product of divisors of the base, or of small primes. In all other case, the model prediction is the largest correct prediction (i.e. l such that $f(l) = l$) that divides k. The list of correct predictions $\mathcal{L}$ varies with the encoding base B. For instance, for $B = 10$, after 300 million examples, the model correctly predicts $\mathcal{L} = \{1, 2, 4, 5, 8, 10, 16, 20, 25, 40, 50, 80, 100...\}$, the GCD of 20 and 30 will be correctly predicted as 10, but the GCD of 210 and 140 will be incorrectly predicted as 10 (instead of 70).

We use these models to generate "dirty" training data $\mathcal{D}(B)$: uniformly sampled pairs of integers (a, b) and their (sometimes incorrect) pseudo-GCD, as generated by a trained transformer using base B. Note: this dataset can be as large as we want. We also create a correct training dataset $\mathcal{C}(B)$, by sampling pairs (a, b) and their correct GCD.

In these experiments, we train models on $\mathcal{D}(B)$ and $\mathcal{C}(B)$, for different values of B. Our goal is to determine whether extensive training on "dirty" data impacts model accuracy.

We focus on 6 bases: $B = 10, 420, 1000, 2017, 2023$ and 4913, after training transformers (on correct GCD) over about 300 millions pairs of integers between one and one million, we achieve the performances listed in Table 1. There, accuracy stands for the proportion of random uniform pairs (a, b) that the model can predict correctly, correct GCD is the number of GCD under 100 that the model correctly predicts (i.e. k such that $f(k) = k$), and correct model predictions are the products of numbers in the associated sets. These models are used to generate $\mathcal{D}(B)$.

In these experiments, all models have four layers, 512 dimensions and 8 attention heads. We consider two architectures: an encoder-only model (17.2M parameters), and an encoder-decoder model (38.7M parameters). The encoder-only model has 2.25 times less parameters, trains twice as fast, and incurs no performance penalty.

Table 1. **Initial performances.** 4-layer transformers trained to predict GCD, on 300 million examples. Our *test* set only contains GCD up to 100, and accuracy is computed on a reweighted test with equal occurance of each GCD. Thus, the Correct GCD lists all those that can be formed from the correct predictions by forming products across the sets (within the first 100 GCD). We freeze the 0th generation model at this stage and use its prediction to generate synthetic data. For each GCD outside the set of its correct predictions, the model will predict the largest GCD it has learned that divides the ground truth.

Base	Accuracy	Correct GCD	Correct predictions
10	85	13	{1,2,4,8,16} {1,5,25}
420	97	38	{1,2,4,8,16}{1,3,9}{1,5,25}{1,7}
1000	94	22	{1,2,4,8,16} {1,5,25}{1,3}
2017	85	4	{1,2}{1,3}
2023	91	16	{1,2,4}{1,3}{1,7}{1,17}
4913	93	17	{1,2,4}{1,3}{1,5}{1,17}

We then train new models (with the same architecture) to predict GCD, from AI data (generated by the above model), and compare to training with correct data – from correct computation of the GCD. When trained on small number of examples (less than 100 million), models learning from AI data achieve better accuracy (Table 2). We believe this is due to the fact that AI data smoothes away all the hard case, therefore presenting the model with a cleaner signal in the initial stages.

Table 2. **Correctly predicted GCD after 30, 60 and 90 million examples.** Dirty and correct datasets.

	30M examples		60M examples		90M examples	
Base	AI	Correct	AI	Correct	AI	Correct
10	13	13	13	13	13	13
420	34	34	38	34	38	35
1000	17	13	22	13	22	14
2017	4	2	4	2	4	4
2023	6	6	11	6	11	6
4913	6	4	7	7	7	7

This pattern changes after extensive training. Table 3 compares performance of models trained on $300M$ and 1 billion

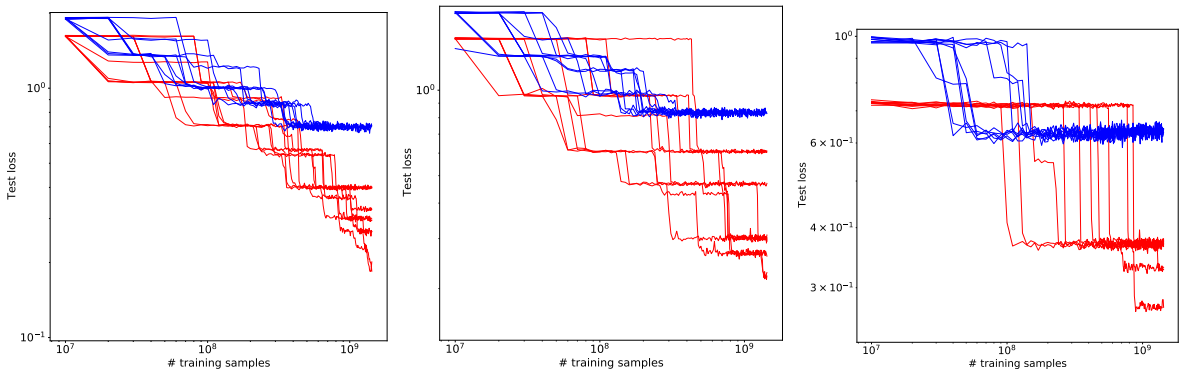

Figure 28. **Test loss for GCD learning.** Test loss of 10 models trained on clean and generated data. From left to right: base 4913, 2023, 1000. Models trained on clean data (red) continune to learn (decreasing test loss) while models trained on AI generated data (blue) stops learning.

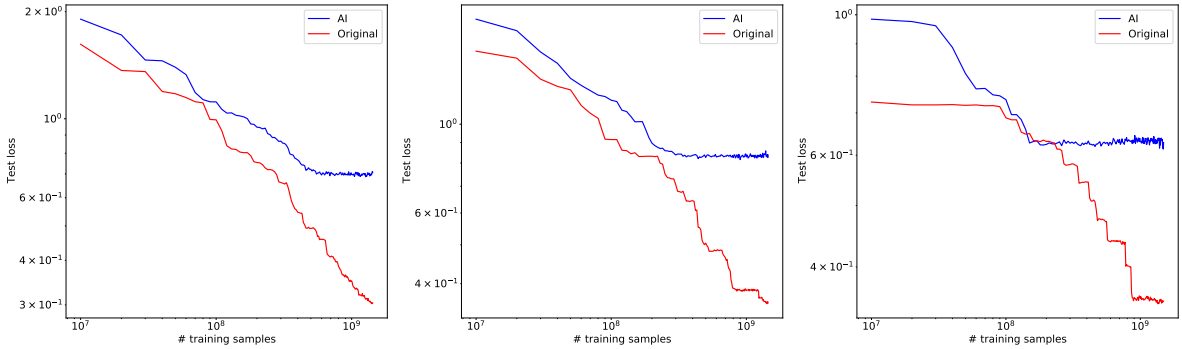

Figure 29. **Average test loss for GCD learning.** Averaged over 10 models trained on clean and generated data. From left to right: base 4913, 2023, 1000.

examples. For all bases B, models trained on $\mathcal{C}(B)$ learn new GCD as training proceeds, whereas models learned on $\mathcal{D}(B)$ never learn beyond their original performance.

Table 3. **Correctly Predicted GCD after** $300M$ **and** 1 **Billion Examples.** AI and correct datasets.

	300M examples		1B examples	
Base	AI	Correct	AI	Correct
10	13	14	13	31
420	38	38	38	40
1000	22	25	22	33
2017	4	6	4	9
2023	16	16	16	32
4913	17	16	17	31

Figures 28 and 29 show that we get the picture predicted by theory: the dirty model learns (until about 300M examples) and then stops learning (while the clean model continues) - its scaling law tapers off as predicted in Theorem 2.1. All the skills the clean model learns after this point are skills the model trained on synthesized data cannot learn (see Figure 30 showing when new learned groups of GCD emerge, and Figure 31 for the learning curve of two models, one trained on the original data, the other on AI data).

Mixing and Grokking We now proceed to train our model on randomly mixed clean and synthesized data for various mixture rates. We train with mixtures of clean and dirty data for mixture fractions of $9\%, 27\%, 50\%$ and 73% of AI-generated data, for bases 1000, 2023 and 4913, to see the grokking effect. Figure 32 illustrates the results. We can see that even for the

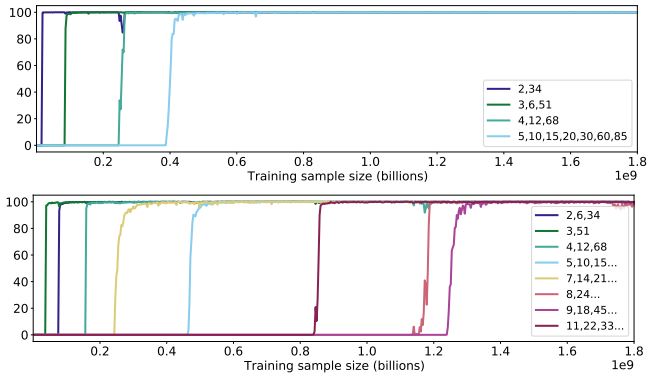

Figure 30. **Emergence of skills (groups of GCDs learned together).** Original (bottom) and AI-synthesized data (top). Base 4913. 1 model for clean/AI data, respectively.

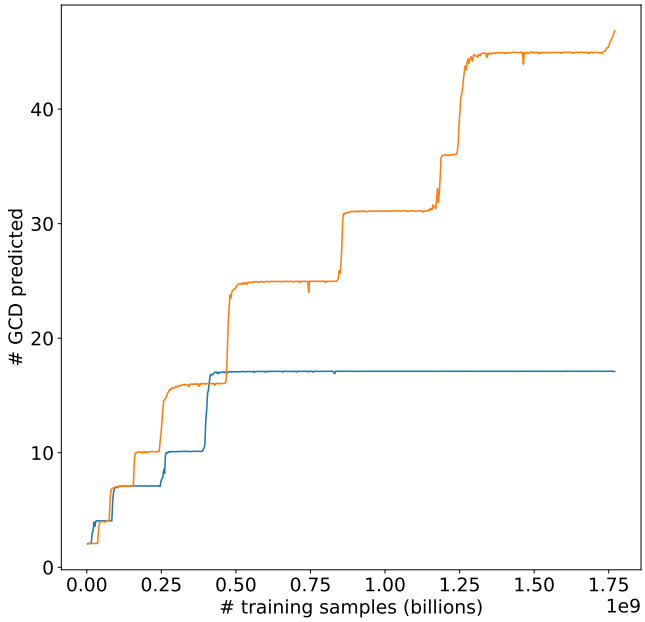

Figure 31. **Learning the GCD.** Learning curve, base 4319. Orange: training on correct GCD. Blue: training on AI generated data.